

Layer-wise Hidden-State Difference Analysis

①

N : normal input

F : false-negative ($\neq N$) attack input

$H_i^{(l)} \in \mathbb{R}^{1 \times d}$: hidden state of token i at layer l

The objective is to identify the layer where the internal representations of the normal and FN inputs diverge most strongly.

Case 1: Token Addition

The newly added token in the attack input has no corresponding token in the normal input. Thus compare only the aligned original tokens:

$$\Delta H_i^{(l)} = H_{i, \text{aligned}}^{F, (l)} - H_i^{N, (l)}$$

where $\Delta H_i^{(l)} \in \mathbb{R}^{1 \times d}$

H^N was denoted as $X^{(N)}$ in my GPT math. (Page 5) Pdf
which is the output of Transformer decoder block.

Case 2: Token replacement

Since the shape of both normal input and attack input is the same, direct subtraction is valid:

$$\Delta H^{(l)} = H^{F, (l)} - H^{N, (l)}$$

(2)

Layer-level score

$$D_l = \frac{1}{T} \sum_{i=1}^T \|H_i^{F(l)} - H_i^{N(l)}\|_2$$

Then the layer where the normal and attack representations differ most:

$$l^* = \arg \max_l D_l$$

(3)

Gradient Attribution for Token-Level FN Analysis (Gradient based Token Attribution)

Input sequence: $(x_1, x_2, \dots, x_T)$

After the forward pass, $H^{(L)} \in \mathbb{R}^{T \times d}$

The prediction at position t is the distribution for the next

token: $Z_T = H_{T,:}^{(L)} W_{LM} \in \mathbb{R}^{1 \times v}$

$$L = X^{(N)} W_{LM}$$

Z was denoted as L in my GPT math.pdf page 5.

Score: $S = Z_T y$ for example $Z_T = [2.1, 0.3, 4.7, 1.2, -0.5]$
then $y=3$ therefore $S=4.7$

If one-hot vector Y_T is used, such as $Y_T = [0, 0, 1, 0, 0]$

where 1 is at position y .

$$S = Z_T Y_T^T$$

$$ds = d(Z_T Y_T^T) = dZ_T Y_T^T + Z_T dY_T^T \quad \text{0 because } Y_T^T \text{ is constant}$$

$$ds = dZ_T Y_T^T \quad (1 \times 1) \rightarrow a = \text{tr}(a)$$

$$ds = \text{tr}(dZ_T Y_T^T) = \text{tr}(Y_T^T dZ_T) \leftarrow \text{tr}(AB) = \text{tr}(BA)$$

$$ds = \text{tr}(G_Z^T dZ_T) \quad \therefore G_Z^T = Y_T^T \rightarrow \boxed{G_Z = Y_T}$$

$$(G_H)_{T,:} = (G_Z)_{T,:} W_{LM}^T$$

$$\boxed{(G_H)_{T,:} = Y_T W_{LM}^T}$$

$$\begin{cases} (G_Z)_{T,:} = Y_T \\ (G_Z)_{t,:} = 0 \quad t \neq T \end{cases}$$

From this point onward, the backpropagation to the original input X is identical to that of tinyGPT.

After all the backpropagations, we get $\text{Gr}_x = \frac{\partial S}{\partial x} \in \mathbb{R}^{T \times d}$ ④

Attribution of token j $a_j = \sum_{k=1}^d x_{jk} \frac{\partial S}{\partial x_{jk}} \leftarrow \text{input } x \text{ gradient}$

$$A = \text{diag}(X \text{Gr}_x^T)$$

Intuitive meaning: Token j 's attribution measures how much $\text{token}(j)$, through all of its embedding dimensions, contributes to the model's output score(S).

A large positive a_j means the token tends to push the score upward, while a large negative a_j means it tends to push the score downward.

Integrated Gradient (IG) based Token Attribution

⑤

$$IG_j = \int_{x_j'}^{x_j} \frac{\partial S(t)}{\partial t_j} dt_j$$

Definition of IG_j :

The net amount contributed by the j -th token change in $S(x)$ as its value increases from x_j' to x_j along the overall input transition; that is, how much of the total change $S(x)$ was generated by the variation of the j -th token while the entire shifts from x' to x (i.e., its attribution).

Substitute $t(\alpha) = x' + \alpha(x - x')$, $t_j(\alpha) = x_j' + \alpha(x_j - x_j')$
 $\frac{dt_j}{d\alpha} = x_j - x_j' \rightarrow dt_j = (x_j - x_j') d\alpha$

$$t_j = x_j' \rightarrow x_j' = x_j' + \alpha(x_j - x_j') \rightarrow \alpha = 0$$

$$t_j = x_j \rightarrow x_j = x_j' + \alpha(x_j - x_j')$$

$$x_j = x_j' + \alpha x_j - \alpha x_j'$$

$$x_j - x_j' = \alpha x_j - \alpha x_j'$$

$$(x_j - x_j') = \alpha (x_j - x_j') \rightarrow \alpha = 1$$

$$IG_j = \int_0^1 \frac{\partial S(t)}{\partial t_j} \bigg|_{t=x'+\alpha(x-x')} (x_j - x_j') d\alpha$$

$$IG_j = (x_j - x_j') \int_0^1 \frac{\partial S(t)}{\partial t_j} \bigg|_{t=x'+\alpha(x-x')} d\alpha$$

Attention Analysis

⑥

For decision point t , which is the last token in our example to decide which MCP tool to choose, inspect the attention row:

$$\text{row: } A[t, :]$$

The position with the largest weight is:

$$j^* = \operatorname{argmax}_j A[t, j]$$

Because GPT uses a causal mask: $A[t, j] = 0$ for $j > t$
So the model can only attend to the current and previous positions.

Normal vs. FN

$$\text{Difference } \Delta A[t, :] = A_{\text{FN}}[t, :] - A_{\text{normal}}[t, :]$$

A large positive $\Delta A[t, j]$ means the FN case attends more strongly to token j .

Hidden State Difference $\rightarrow$ Where does the model's internal representation change?

Attention $\rightarrow$ At decision point t , which tokens receive attention?

$| \mathbf{e} \rightarrow$ which input contributes to the score?

Occlusion $\rightarrow$ What changes when a token is removed?